Description of Data Input File

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The pore-scale network model, `pnflowPy`, is written in `Python` and uses a keyword-based input file. Its format is similar to that of the pnflow code written in `C++`, but not all `pnflow` keywords are currently supported in `pnflowPy`. Some of the keywords are optional and the order of listing is not important. Comments in the data file are indicated by `"%"`, implying that the rest of the line will be discarded. All keywords should be terminated by `"#"`. The input data file can be supplied as an argument to the executable.

```
C:\path _to_executable\pnflowPy .exe input_file.dat
```

1 Required keywords

1.1 NETWORK/TITLE

The keyword `NETWORK/TITLE` specifies the files containing data such as inscribed radii, volumes and the connection list defining the network topology. The network data is provided through four ASCII files which must share the same base name (e.g., filename) and follow the naming convention: `filename_node1.dat`, `filename_node2.dat`, `filename_link1.dat` and `filename_link2.dat`. Only the prefix `filename` is to be specified. All output files are prefixed by the indicated title. If this keyword is omitted, the title is taken to be the same as the name of the network.

```
TITLE Bentheimer;
NETWORK F Bentheimer;
```

1.2 LOAD_INIT_NETWORK_STATE

1. Whether to load an initial network state from a previous simulation or not for the quasi-static simulation. Acceptable input is 'T' or 'F'.

2. The path to the initial network state from where simulation should commence. A .pkl file is expected.

The keyword `LOAD_INIT_NETWORK_STATE` specifies the starting point of the quasi-static simulation. If this keyword is missing or commented out or the first parameter is 'F', the simulation starts from a single phase.

```
LOAD_INIT_NETWORK_STATE        T        path/to/the/network/state/pkl/file ;
```

1.3 LOAD_INIT_RIPENING_STATE

1. Whether to load an initial network state from a previous simulation or not for the ripening simulation. Acceptable input is 'T' or 'F'.

2. The path to the initial network state from where simulation should commence. A .pkl file is expected.

```
TITLE Bentheimer;     % Base name for the output files

NETWORK F Bentheimer; % Base name of the network files, without _link1.dat,
                      % _link2.dat, node1.dat and node2.dat

LOAD_INIT_NETWORK_STATE        T        path/to/the/network/state/pkl/file ;
LOAD_INIT_RIPENING_STATE      T        path/to/the/network/ripening/state/pkl/file;

SAT_CONTROL:
%finalSat   maxPc    deltaSw   minDeltaPc   deltaPcFraction
%-------------------------------------------------------------------------
   0.0       1e6       0.1      1000.0           0.15
   1.00     -1000      0.1       100.0           0.15
   0.0       2400      0.1      1000.0           0.15

CALC_BOX  0.15 1.0; % X bounds of the network used in rel-perms calculations

% Morrow's model parameters
%             Model    min     max    del     eta         RCrl    SepAng
%-------------------------------------------------------------------------
INIT_CONT_ANG:  1       0       0     -0.2    -3.0        rand     25.0;

%             Model    min     max    del     eta         RCrl    SepAng
%-------------------------------------------------------------------------
EQUIL_CON_ANG: 3       45      45      0.2     3.0        rand     25.0;

RES_DIR         path/to/save/result/files ; % relative to the project dir

PORE_FILL_WGT     15000.0  15000.0  15000.0  15000.0  15000.0  15000.0;

% Fluid properties:
%      interfacial    Water    Oil     Water       Oil
%        tension       visc    visc    density     density
%        (mN/m)        (cp)    (cp)    (kg/m3)     (kg/m3)
%-------------------------------------------------------------------------
FLUID    57.0          0.821   0.838     1000.0      1.98;     %CO2

SAT_CONVERGENCE
%  minNumFillings   initStepSize   cutBack   maxIncr   stable disp
%-------------------------------------------------------------------------
          50              0.1          0.8       2.0          F ;

RAND_SEED        1099002   ;

% Ostwald ripening parameters:
RIPENING
% mode  alpha  duration   dt      D       P   T       H      moles_tol   pc_tol  max_iter
%                (s)    (s)   (m2/s)  (Pa) (K) (mol/m3/pa)  (mols)      (Pa)
   T    0.617    1200    1.5  4.89e-9  1e6  298   7.8e-6     1e-17      1e-4       50;
```

Figure 1: Sample input file for the pnflowPy two-phase flow simulations explaining the important keywords.

The keyword `LOAD_INIT_RIPENING_STATE` specifies the starting point of the ripening simulation. If this keyword is missing or commented out or the first parameter is 'F', the ripening simulation starts from time, $t = 0$ s.

```
LOAD_INIT_RIPENING_STATE      T        path/to/the/network/ripening/state/pkl/file;
```

1.4 SAT_CONTROL

Each line represents a separate flooding cycle such as primary oil flooding or secondary water flooding.

1. Final water saturation target after the flooding cycle (fraction). The actual value might not actually be reached due to trapping or water retained in corners.
2. The cycle can also be terminated when a capillary pressure (Pa) is reached.
3. Target water saturation interval (fraction) between reporting results. The state of the model will be evaluated at the first possible configuration after the incremental target has been reached.
4. Target capillary pressure interval (Pa) between reporting results.
5. Fraction used to determine next reporting capillary pressure.

```
SAT_CONTROL:
%finalSat   maxPc     deltaSw    minDeltaPc    deltaPcFraction
%-----------------------------------------------------------------------
   0.0        1e6        0.1       1000.0            0.15
   1.00      -1000       0.1        100.0            0.15
   0.0        2400       0.1       1000.0            0.15
```

1.5 CALC_BOX

Most of the elements close to the injection faces will be filled by the injecting fluid, resulting in most of the pressure loss occurring in this region when solving the pressure field for the displaced fluid. To avoid these boundary effects it is normal to only use a fraction of the network (away from the injecting faces) when calculating relative permeabilities.

1. Dimensionless location for lower boundary.
2. Dimensionless location for higher boundary.

```
CALC_BOX  0.15 1.0;
```

1.6 INIT_CONT_ANG/EQUIL_CON_ANG

The keyword `INIT_CONT_ANG` is used to assign the contact angles for the primary drainage cycle and `EQUIL_CON_ANG` is used to assign the contact angles for the second (water-injection) cycle, which can be different from the primary drainage due to wettability alteration. The keywords `INIT_CONT_ANG` and `EQUIL_CON_ANG` assign the contact angles using a random or Weibull distribution.

1. Contact angle hysteresis model which can be 1, 2, or 3, indicating Morrow's contact angle hysteresis models 1 to 3.
2. The smallest contact angle.
3. The largest contact angle.
4. Weibull coefficient δ.
5. Weibull coefficient η.

6. The contact angle correlation with pore size which can be `rMin`, `rMax` or `rand`. `rMin` implies smaller contact angles are assigned to smaller pores, while the `rMax` option leads to the opposite, and `rand` means no correlation to pore size.

7. Default separation angle (`25 degrees`) (applicable when the contact angle hysteresis model is 2).

If the given Weibull distribution coefficients are negative, a uniform distribution between the min and max contact angles will be used instead. Otherwise, the two Weibull coefficients δ and η describe the contact angle distribution using the truncated Weibull distribution equation given as:

$$\theta = \theta_{\min} + (\theta_{\max} - \theta_{\min}) \left(-\delta \ln \left[x(1 - e^{-1/\delta}) + e^{-1/\delta} \right] \right)^{1/\eta} \tag{1}$$

where x is a random number between 0 and 1.

```
%Morrow's model parameters
%             Model     min      max     del      eta          RCrl     SepAng
%-----------------------------------------------------------------------
INIT_CONT_ANG:  1        0        0      -0.2     -3.0         rand      25.0;
EQUIL_CON_ANG:  3        45       45       0.2      3.0         rand      25.0;
```

1.7 RES_DIR

The keyword `RES_DIR` is used to state the directory where the result files should be saved.

```
RES_DIR        path/to/save/result/files ;
```

1.8 PORE_FILL_WGT

The pore filling algorithm is "Blunt2", other algorithm could be implemented later. The keyword `PORE_FILL_WGT` is used to state the filling weights when the cooperative pore filling, I_n is I_0, I_1, I_2, I_3, I_4 and I_5 respectively where n is the number of throats connected to the pore that contain non-wetting fluid.

```
PORE_FILL_WGT     0.0  15000.0  15000.0  15000.0  15000.0  15000.0;
```

1.9 FLUID

The fluid description is very simple with no pressure-dependent properties.

1. Interfacial tension ($\mathrm{mN \cdot m^{-1}}$)

2. Water viscosity (cP)

3. Oil viscosity (cP)

4. Water density ($\mathrm{kg \cdot m^{-3}}$)

5. Oil density ($\mathrm{kg \cdot m^{-3}}$)

```
% Fluid properties:
%      interfacial     Water    Oil      Water       Oil
%        tension        visc    visc     density     density
%        (mN/m)         (cp)    (cp)     (kg/m3)     (kg/m3)
%-----------------------------------------------------------------------
FLUID    57.0          0.821   0.838     1000.0       1.98;     %CO2
```

1.10 SAT_CONVERGENCE

This keyword controls the accuracy at which the incremental water saturation target is reached. It is generally not feasible to calculate water saturation after each filling event, as that would be computationally very expensive. When starting a new saturation step, an approximate number of required filling events is estimated. This estimate is however quite uncertain as the water saturation does not vary linearly with the number of filling events. The actual number of filling events performed before recalculating water saturation should therefore be a relatively small fraction of the initial estimate. Subsequently an updated estimate of the required number of filling events is made, using information from the previous step. Still, only a fraction of the estimated filling events should be performed before recalculating water saturation, as the estimate remains fairly uncertain.

1. Minimum number of filling events between calculating water saturation.
2. Fraction applied to the initial estimate of required number of filling events.
3. Fraction applied to subsequent estimates of required number of filling events.
4. Maximum increase factor in required number of filling events estimates.
5. Only solve for relative permeability when a stable capillary configuration is reached (true 'T' or false 'F')?

```
SAT_CONVERGENCE
%  minNumFillings   initStepSize   cutBack   maxIncr   stable disp
%----------------------------------------------------------------------
          50              0.1          0.8       2.0          F ;
```

1.11 RAND_SEED

This is the seed to the random number generator, which should be a large positive integer. If the keyword is omitted, the computer clock will be used as the seed.

```
RAND_SEED        1099002   ;
```

1.12 RIPENING

The keyword **RIPENING** is used to specify the parameters to be used during the ripening simulation:

1. The ripening mode whether it is transient ('T') or not ('F'), that is equilibrium.
2. Alpha parameter for tuning the capillary pressures in the gas ganglia, values in the range [0, 1].
3. Duration for the ripening simulation, in seconds.
4. Maximum time steps at which update is done during ripening, in seconds.
5. Diffusivity coefficient, in square meter per seconds.
6. Imposed pressure, in pascal.
7. System temperature, in kelvin.
8. Henry's constant, in moles per unit cubic meter per unit pascal.
9. Moles absolute tolerance, in moles.
10. Capillary pressure absolute tolerance, in pascal.
11. Maximum iteration at any instant during ripening.

```
% Ostwald ripening parameters:
RIPENING
% mode  alpha  duration   dt     D        P    T       H       moles_tol   pc_tol  max_iter
%                (s)     (s)   (m2/s)  (Pa) (K) (mol/m3/pa)  (mols)      (Pa)
   T    0.617    1200    1.5  4.89e-9  1e6  298   7.8e-6     1e-17      1e-4       50;
```